Project :Dhaka Network

CSE421

Section: 20

Group Members:

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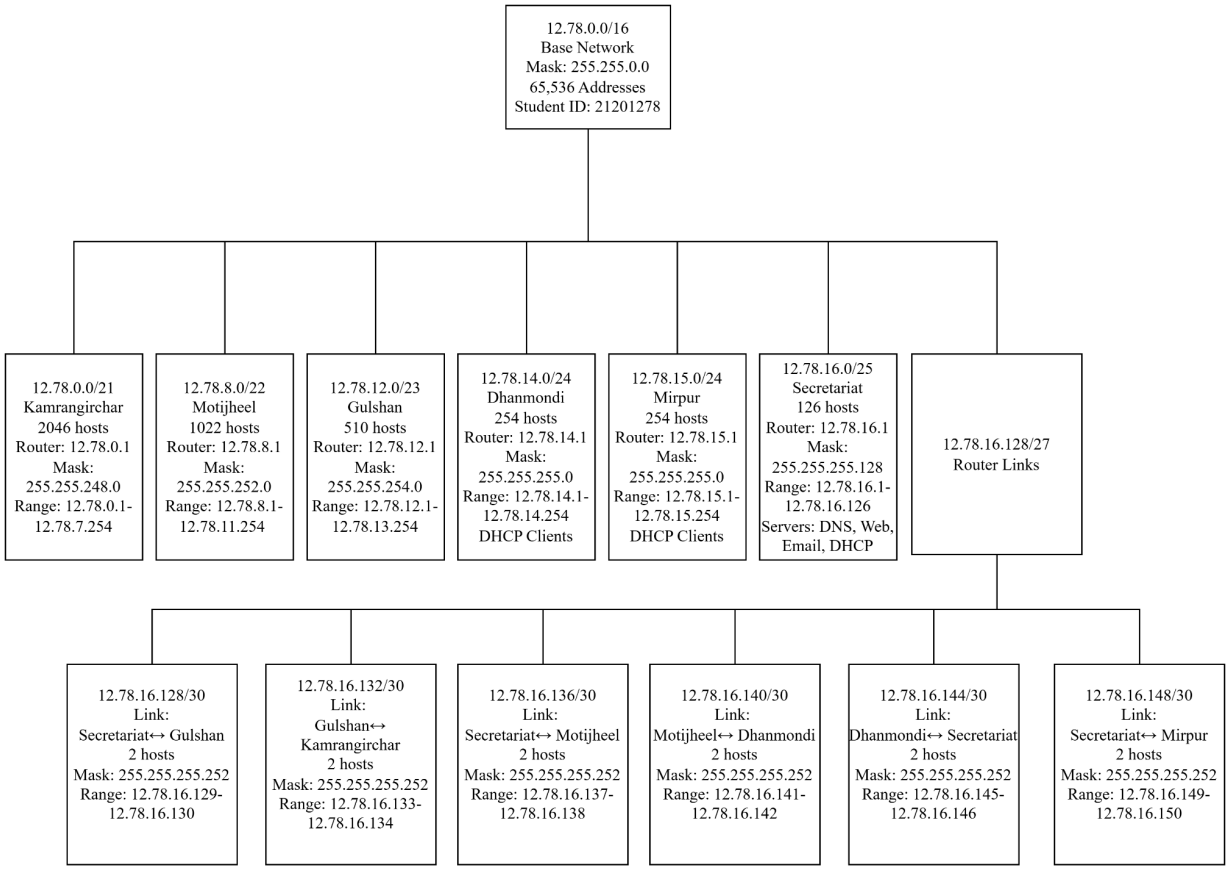
Project Overview

This project implements a secure enterprise network connecting six major areas of Dhaka City using Cisco Packet Tracer. The network enables efficient communication for government, commerce, and education sectors.
Selected ID - 21201278

Base network address is 12.78.0.0/16.

VLSM Tree



IP Addressing Table

Location	Server	Network Address	IP Address	Subnet Mask	Default Gateway	Hosts Needed
Kamrangirchar LAN	Subnet	12.78.0.0/21	-	255.255.248.0	-	2046 hosts
	Router G0/0	-	12.78.0.1	255.255.248.0	-	LAN Gateway
Motijheel LAN	Subnet	12.78.8.0/22	-	255.255.252.0	-	1024 hosts
	Router G0/0	-	12.78.8.1	255.255.252.0	-	LAN Gateway
Gulshan LAN	Subnet	12.78.12.0/23	-	255.255.254.0	-	510 hosts
	Router G0/0	-	12.78.12.1	255.255.254.0	-	LAN Gateway
Dhanmondi LAN	Subnet	12.78.14.0/24	-	255.255.255.0	-	DHCP Clients
	Router G0/0	-	12.78.14.1	255.255.255.0	-	LAN Gateway
	DHCP Scope	-	12.78.14.100–200	-	12.78.14.1	DHCP Pool
Mirpur LAN	Subnet	12.78.15.0/24	-	255.255.255.0	-	DHCP Clients
	Router G0/0	-	12.78.15.1	255.255.255.0	-	LAN Gateway
	DHCP Scope	-	12.78.15.100–200	-	12.78.15.1	DHCP Pool
Secretariat LAN	Subnet	12.78.16.0/25	-	255.255.255.128	-	Servers
	Router G0/0	-	12.78.16.1	255.255.255.128	-	LAN Gateway
	DNS Server	-	12.78.16.10	255.255.255.128	12.78.16.1	Service
	Web Server	-	12.78.16.20	255.255.255.128	12.78.16.1	Service
	DHCP Server	-	12.78.16.30	255.255.255.128	12.78.16.1	Service
	Email Server	-	12.78.16.40	255.255.255.128	12.78.16.1	Service

Router Links	Secretariat-Gulshan	12.78.16.128/30	12.78.16.129 (Secretariat)	255.255.255.252	-	Point-to-point
			12.78.16.130 (Gulshan)	255.255.255.252	-	
	Gulshan-Kamrangirchar	12.78.16.132/30	12.78.16.133 (Gulshan)	255.255.255.252	-	
			12.78.16.134 (Kamrangirchar)	255.255.255.252	-	
	Secretariat-Motijheel	12.78.16.136/30	12.78.16.137 (Secretariat)	255.255.255.252	-	Main Link
			12.78.16.138 (Motijheel)	255.255.255.252	-	
	Motijheel-Dhanmondi	12.78.16.140/30	12.78.16.141 (Motijheel)	255.255.255.252	-	
			12.78.16.142 (Dhanmondi)	255.255.255.252	-	
	Dhanmondi-Secretariat	12.78.16.144/30	12.78.16.145 (Dhanmondi)	255.255.255.252	-	Backup Link
			12.78.16.146 (Secretariat)	255.255.255.252	-	
	Secretariat-Mirpur	12.78.16.148/30	12.78.16.149 (Secretariat)	255.255.255.252	-	
			12.78.16.150 (Mirpur)	255.255.255.252	-	
Switch Management	Sw_Secretariat	-	12.78.16.200	255.255.255.128	12.78.16.1	Mgmt VLAN
	Sw_Kamrangirchar	-	12.78.0.200	255.255.248.0	12.78.0.1	Mgmt VLAN
	Sw_Motijheel	-	12.78.8.200	255.255.252.0	12.78.8.1	Mgmt VLAN
	Sw_Gulshan	-	12.78.12.200	255.255.254.0	12.78.12.1	Mgmt VLAN
	Sw_Dhanmondi	-	12.78.14.200	255.255.255.0	12.78.14.1	Mgmt VLAN
	Sw_Mirpur	-	12.78.15.200	255.255.255.0	12.78.15.1	Mgmt VLAN

Commands

R_Secretariat Router

```
Router>en
Router#conf t
Router(config)#hostname R_Secretariat

R_Secretariat(config)#int g0/0
R_Secretariat(config-if)#ip address 12.78.16.1 255.255.255.128
R_Secretariat(config-if)#no shutdown
```

To Gulshan

```
R_Secretariat>en
R_Secretariat#conf t
R_Secretariat(config)#int s0/0/0
R_Secretariat(config-if)#ip address 12.78.20.1 255.255.255.252
R_Secretariat(config-if)#clock rate 64000
R_Secretariat(config-if)#no shut
```

To Motijheel

```
R_Secretariat(config-if)#int s0/0/1
R_Secretariat(config-if)#ip address 12.78.20.9 255.255.255.252
R_Secretariat(config-if)#clock rate 64000
R_Secretariat(config-if)#no shut
```

To Dhanmondi

```
R_Secretariat(config-if)#int s0/1/0
```

```
R_Secretariat(config-if)#ip address 12.78.20.17 255.255.255.252
R_Secretariat(config-if)#clock rate 64000
R_Secretariat(config-if)#no shut
```

To Mirpur

```
R_Secretariat(config-if)#int s0/1/1
R_Secretariat(config-if)#ip address 12.78.20.21 255.255.255.252
R_Secretariat(config-if)#clock rate 64000
R_Secretariat(config-if)#no shut
```

Recursive static route to Gulshan

```
R_Secretariat(config-if)#ip route 12.78.12.0 255.255.254.0 12.78.20.2
```

RIP

```
R_Secretariat(config)#router rip
R_Secretariat(config-router)#version 2
R_Secretariat(config-router)#network 12.0.0.0
R_Secretariat(config-router)#no auto-summary
R_Secretariat(config-router)#exit
```

R_Gulshan Router

```
Router>en
Router#conf t
Router(config)#hostname R_Gulshan

R_Gulshan(config)#int g0/0
R_Gulshan(config-if)#ip address 12.78.12.1 255.255.254.0
R_Gulshan(config-if)#no shutdown
R_Gulshan(config-if)#exit

R_Gulshan(config)#int s0/0/0
R_Gulshan(config-if)#ip address 12.78.20.2 255.255.255.252
R_Gulshan(config-if)#no shutdown
R_Gulshan(config-if)#exit

R_Gulshan(config)#int s0/0/1
R_Gulshan(config-if)#ip address 12.78.20.5 255.255.255.252
R_Gulshan(config-if)#clock rate 64000
```

```
R_Gulshan(config-if)#no shutdown
R_Gulshan(config-if)#exit

R_Gulshan(config)#ip route 12.78.16.0 255.255.255.128 12.78.20.1
R_Gulshan(config)#ip route 12.78.0.0 255.255.248.0 s0/0/1

R_Gulshan(config)#router rip
R_Gulshan(config-router)#version 2
R_Gulshan(config-router)#network 12.0.0.0
R_Gulshan(config-router)#no auto-summary
R_Gulshan(config-router)#exit
```

R_Kamrangirchar Router

```
Router>en
Router#conf t
Router(config)#hostname R_Kamrangirchar

R_Kamrangirchar(config)#int g0/0
R_Kamrangirchar(config-if)#ip address 12.78.0.1 255.255.248.0
R_Kamrangirchar(config-if)#no shutdown
R_Kamrangirchar(config-if)#exit

R_Kamrangirchar(config)#int s0/0/0
R_Kamrangirchar(config-if)#ip address 12.78.20.6 255.255.255.252
R_Kamrangirchar(config-if)#no shutdown
R_Kamrangirchar(config-if)#exit

R_Kamrangirchar(config)#ip route 12.78.12.0 255.255.254.0 s0/0/0

R_Kamrangirchar(config)#router rip
R_Kamrangirchar(config-router)#version 2
R_Kamrangirchar(config-router)#network 12.0.0.0
R_Kamrangirchar(config-router)#no auto-summary
R_Kamrangirchar(config-router)#exit
```

R_Motijheel Router

```
Router>en
```

```
Router#conf t
Router(config)#hostname R_Motijheel

R_Motijheel(config)#int g0/0
R_Motijheel(config-if)#ip address 12.78.8.1 255.255.252.0
R_Motijheel(config-if)#no shutdown
R_Motijheel(config-if)#exit

R_Motijheel(config)#int s0/0/0
R_Motijheel(config-if)#ip address 12.78.20.10 255.255.255.252
R_Motijheel(config-if)#no shutdown
R_Motijheel(config-if)#exit

R_Motijheel(config)#int s0/0/1
R_Motijheel(config-if)#ip address 12.78.20.13 255.255.255.252
R_Motijheel(config-if)#clock rate 64000
R_Motijheel(config-if)#no shutdown
R_Motijheel(config-if)#exit

R_Motijheel(config)#ip route 12.78.16.0 255.255.255.128 12.78.20.14 200

R_Motijheel(config)#router rip
R_Motijheel(config-router)#version 2
R_Motijheel(config-router)#network 12.0.0.0
R_Motijheel(config-router)#no auto-summary
R_Motijheel(config-router)#exit
```

R_Dhanmondi Router

```
Router>en
Router#conf t
Router(config)#hostname R_Dhanmondi

R_Dhanmondi(config)#int g0/0
R_Dhanmondi(config-if)#ip address 12.78.14.1 255.255.255.0
R_Dhanmondi(config-if)#ip helper-address 12.78.16.8
R_Dhanmondi(config-if)#no shutdown
R_Dhanmondi(config-if)#exit

R_Dhanmondi(config)#int s0/0/0
R_Dhanmondi(config-if)#ip address 12.78.20.14 255.255.255.252
R_Dhanmondi(config-if)#no shutdown
```

```
R_Dhanmondi(config-if)#exit

R_Dhanmondi(config)#int s0/0/1
R_Dhanmondi(config-if)#ip address 12.78.20.18 255.255.255.252
R_Dhanmondi(config-if)#no shutdown
R_Dhanmondi(config-if)#exit

R_Dhanmondi(config)#router rip
R_Dhanmondi(config-router)#version 2
R_Dhanmondi(config-router)#network 12.0.0.0
R_Dhanmondi(config-router)#no auto-summary
R_Dhanmondi(config-router)#exit
```

R_Mirpur Router

```
Router>en
Router#conf t
Router(config)#hostname R_Mirpur

R_Mirpur(config)#int g0/0
R_Mirpur(config-if)#ip address 12.78.15.1 255.255.255.0
R_Mirpur(config-if)#ip helper-address 12.78.16.8
R_Mirpur(config-if)#no shutdown
R_Mirpur(config-if)#exit

R_Mirpur(config)#int s0/0/0
R_Mirpur(config-if)#ip address 12.78.20.22 255.255.255.252
R_Mirpur(config-if)#no shutdown
R_Mirpur(config-if)#exit

R_Mirpur(config)#router rip
R_Mirpur(config-router)#version 2
R_Mirpur(config-router)#network 12.0.0.0
R_Mirpur(config-router)#no auto-summary
R_Mirpur(config-router)#exit
```

DHCP Relay for Mirpur and Dhanmondi

```
R_Secretariat(config-if)#int g0/0
R_Secretariat(config-if)#ip helper-address 12.78.16.10
R_Secretariat(config-if)#exit
```

```
R_Secretariat(config)#end
write memory
```

Assumptions Made in the Network Design

Routing Assumptions:

- RIP Version: Used RIPv2 as specified (version 2 with no auto-summary)
- Administrative Distance:
    - Floating static route AD = 200 (higher than RIP's 120)
    - Default static route AD = 1
- Route Redistribution: Assumed no route redistribution needed between static and RIP

Router Assumptions:

- Interface Types:
    - Serial interfaces for WAN links
    - GigabitEthernet for LAN connections
- Clock Rate: Assumed 64000 bps for all serial DCE interfaces

DHCP Service:

- DHCP Server Location: Separate DHCP server device (not router-based)
- DHCP Scope:
    - Mirpur: 12.78.15.11-254
    - Dhanmondi: 12.78.14.11-254
    - Excluded .1-.10 for static devices
- DHCP Options:
    - Default gateway set to router interface

    - DNS server: 12.78.16.5
    - Domain name: dhaka.gov
- DHCP Relay: Used `ip helper-address` on router interfaces